Lecture 7: Hypothesis Testing

STAT 521
Applied Probability and Statistical Inference

Brenda Betancourt

Example: US court trials

A US court considers two possible claims about a defendant: she is either innocent or guilty. If we set these claims up in a hypothesis framework, which would be the null hypothesis and which the alternative?

- We start a trial under the assumption the person is innocent (null hypothesis).
- Jurors examine the evidence to see whether it convincingly shows a defendant is guilty.
- However, even if the jurors leave unconvinced of guilt beyond a reasonable doubt, this does not mean they believe the defendant is innocent.

Hypothesis testing framework

- We start with a *null hypothesis (H_0)* that represents the status quo.
- We then come up with an *alternative hypothesis (H_A)* that represents our research question, i.e. what we're testing for.
- We conduct a hypothesis test under the assumption that the null hypothesis is true.
- We examine the evidence, the *p-value*, to see whether it favors the alternative hypothesis.
- Failing to find strong evidence for the alternative hypothesis is not equivalent to accepting the null hypothesis $\rightarrow$ we say that we fail to reject the null hypothesis.
- If the data provides strong evidence against the null, then we reject the null hypothesis in favor of the alternative.

Decision errors

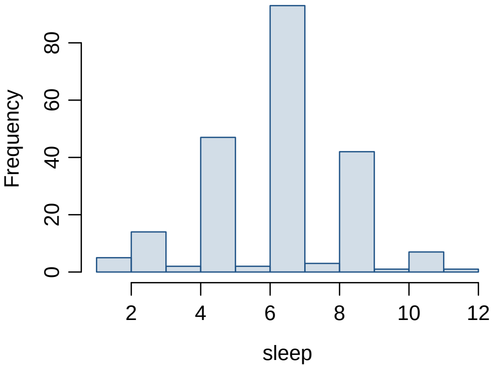
- The type I error is committed if we reject the null hypothesis when it is true *(α)*. α is also called the level of significance of the test.
- The type II error is committed if we fail to reject the null hypothesis when it is false and the alternative hypothesis is true *(β)*.
- There is a tradeoff between the two errors. Is one more acceptable than the other? *Researchers normally choose α to be small i.e. 0.01, 0.05, 0.1.*

Based on a particular significance level α:

If p-value $< \alpha$ we reject H_0
If p-value $> \alpha$ we fail to reject H_0

Hypothesis Test of Student Sleep

A poll by the National Sleep Foundation found that college students average about 7 hours of sleep per night. A sample of 217 GMU students yielded an average of 6.7 hours, with a standard deviation of 2.03 hours. Assuming that this is a random sample representative of all college students, do these data provide convincing evidence that GMU students on average sleep 7 hours per night?



Assumptions & conditions for inference

Before doing inference using this data set, we must make sure that the assumptions & conditions necessary for inference are satisfied:

1. *Random sampling and Independence Assumption:* We can assume that this is a random sample and the hours slept by a student in this sample is independent of the other students.

2. *Nearly Normal Condition:* The distribution appears to be roughly normal and/or $n > 30$ so we can assume that the distribution of the sample means is nearly normal.

Setting the hypotheses

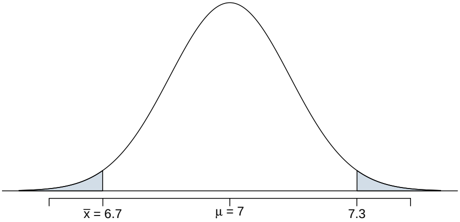
$H_0 : \mu = 7$ (GMU students sleep 7 hours per night on average)

$H_a : \mu \neq 7$ (GMU students do not sleep 7 hours per night on average)

- This is called a two-tailed test $H_0 : \mu = \mu_0$ vs. $H_a : \mu \neq \mu_0$
- If in fact the null hypothesis is true, $\bar{x}$ is distributed nearly normally with mean $\mu = 7$ and standard deviation $\frac{s}{\sqrt{n}} = \frac{2.03}{\sqrt{217}} = 0.14$.
- We would like to find out how likely it is to observe a sample mean at least as far from the data as our current sample mean (6.7), if in fact the null hypothesis is true.

p-values

- The *p-value* is the probability of getting data as extreme or more extreme than what we actually observed, if the null hypothesis was true.



- The shaded area represents the rejection region: values of $\bar{x}$ that support the alternative hypothesis and contradict the null hypothesis.

- The probability of the values in the rejection region is what we call the *p-value*.

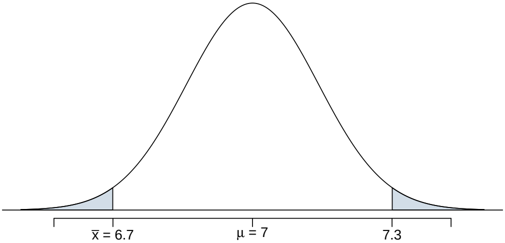
p-values

- If the p-value is *low* (lower than the significance level, α, which is usually 5%) we say that it would be very unlikely to observe the data if the null hypothesis were true, and hence *reject H_0*.

- If the p-value is *high* (higher than α) we say that it is likely to observe the data even if the null hypothesis were true, and hence *do not reject H_0*.

- We never accept H_0 since we're not in the business of trying to prove it. We simply want to know if the data provide convincing evidence to support H_A.

Calculating the p-value



$$\bar{x} \sim N\left(\mu = 7, \frac{2.03}{\sqrt{217}} = 0.14\right)$$

$$Z = \frac{6.7 - 7}{0.14} = -2.14$$

$$\begin{aligned} p - value &= \text{Probability that a sample mean is} \leq 6.7 \text{ or} \geq 7.3 \text{ assuming } \mu = 7 \\ &= P(\bar{x} < 6.7 \text{ or } \bar{x} > 7.3 \mid \mu = 7) \\ &= P(Z < -2.14) + P(Z > 2.14) = 0.0162 + 0.0162 = 0.0324 \end{aligned}$$

Based on a p-value of 0.0324, which of the following is true?
$H_0 : \mu = 7$ (GMU students sleep 7 hours per night on average)
$H_A : \mu \neq 7$ (GMU students do not sleep 7 hours per night on average)

a. Fail to reject H_0, the data provide convincing evidence that GMU students do not sleep 7 hours on average.
b. Reject H_0, the data provide convincing evidence that GMU students do not sleep 7 hours on average.
c. *Reject H_0, the data provide convincing evidence that GMU students do not sleep 7 hours on average.*
d. Reject H_0, the data prove that GMU students sleep 7 hours on average.
e. Fail to reject H_0, the data do not provide convincing evidence that GMU students do not sleep 7 hours on average.
f. Reject H_0, the data provide convincing evidence that GMU students in this sample do not sleep 7 hours on average.

Number of college applications - hypotheses

The same survey asked how many colleges students applied to, and 206 students responded to this question. This sample yielded an average of 9.7 college applications with a standard deviation of 7. College Board website states that counselors recommend students apply to roughly 8 colleges. Which of the following are the correct set of hypotheses to test if these data provide convincing evidence that the average number of colleges all GMU students apply to is *higher* than recommended.

(a) $H_0 : \mu = 9.7$
$H_A : \mu > 9.7$

(b) $H_0 : \mu = 8$
$H_A : \mu > 8$

(c) $H_0 : \mu = 8$
$H_A : \mu > 8$

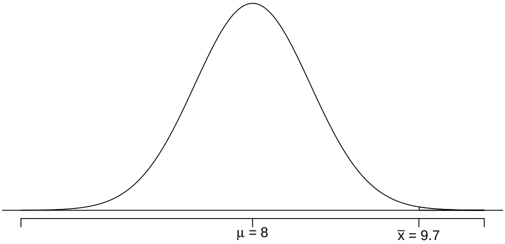
(d) $H_0 : \bar{x} = 8$
$H_A : \bar{x} > 8$

(e) $H_0 : \mu = 8$
$H_A : \mu > 9.7$

This is called a right-tailed test.

http://www.collegeboard.com/student/apply/the-application/151680.html

Number of college applications - test statistic



$$\bar{x} \sim N\left(\mu = 8, \frac{7}{\sqrt{206}} = 0.5\right)$$

$$Z = \frac{9.7 - 8}{0.5} = 3.4$$

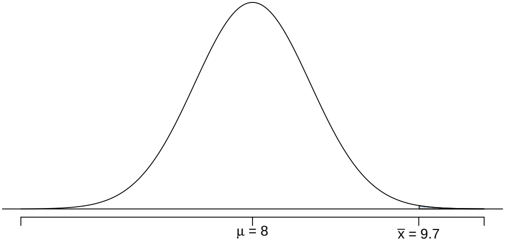
The sample mean is 3.4 standard deviations away from the hypothesized value.

Is this considered unusually (significantly) high?

Yes, but we can quantify how unusual it is using a p-value.

Number of college applications - p-value

p-value: probability of observing data at least as favorable to H_A as our current data set (a sample mean greater than 9.7), if in fact H_0 was true (the true population mean was 8).



$$P(\bar{x} > 9.7 \mid \mu = 8) = P(Z > 3.4) = 0.0003$$

Number of college applications - Making a decision

- p-value = 0.0003
  - If the true average of the number of colleges GMU students applied to is 8, there is only 0.03% chance of observing a random sample of 206 GMU students who on average apply to 9.7 or more schools.
  - This is a pretty low probability for us to think that a sample mean of 9.7 or more schools is likely to happen simply by chance.
- Since p-value is *low* (lower than 5%) we *reject H_0*.
- The data provide convincing evidence that GMU students average apply to more than 8 schools.
- The difference between the null value of 8 schools and observed sample mean of 9.7 schools is *not due to chance* or sampling variability.

Recap: Hypothesis testing framework

1. Set the hypotheses.
2. Check assumptions and conditions.
3. Calculate a *test statistic* and a p-value.
4. Make a decision, and interpret it in context of the research question.

Recap: Hypothesis testing for a population mean

1. Set the hypotheses
   - $H_0 : \mu = null\ value$
   - $H_A : \mu <$ or $>$ or $\neq null\ value$
2. Check assumptions and conditions
   - Independence: random sample/assignment
   - Normality: nearly normal population or $n \geq 30$, no extreme skew
3. Calculate a *test statistic* and a p-value (draw a picture!)

$$Z = \frac{\bar{x} - \mu}{SE}, \ where\ SE = \frac{s}{\sqrt{n}}$$

4. Make a decision, and interpret it in context of the research question
   - If p-value $< \alpha$, reject H_0, data provide evidence for H_A
   - If p-value $> \alpha$, do not reject H_0, data do not provide evidence for H_A

Exercise: Hypothesis testing

A random sample of n $=$ 36 cars driving on a residential street had a sample mean of 38 miles/hour, and a standard deviation of 12 miles per hour.

- Obtain a 95% Confidence Interval for the true population mean speed of all drivers on the street.
- Test whether the population mean speed is different from the posted speed limit of 35 mph with a confidence level of $\alpha = 0.05$.
- Use the 95% Confidence interval to test the hypothesis that the mean speed is different from the posted speed limit of 35.

Relationship between CI and Hypothesis Test

For a two-sided test at significance level α:

$$H_0 : \mu = \mu_0 \quad \text{vs.} \quad H_a : \mu \neq \mu_0$$

- The $(1 - \alpha) \times 100\%$ confidence interval contains all values of μ that *would not be rejected* by a two-sided test at level α.
- Therefore:
  - If μ_0 is *inside* the CI $\Rightarrow$ fail to reject H_0.
  - If μ_0 is *outside* the CI $\Rightarrow$ reject H_0.

Example:
95% CI for mean speed: (33.94, 42.06) mph.
Since 35 mph is inside the interval $\Rightarrow$ fail to reject H_0.

When They May Not Coincide

The CI and hypothesis test results can differ if:

1. Different confidence levels are used (e.g., 90% CI vs. $\alpha = 0.05$ test).
2. The test is **one-sided** but the CI is **two-sided**.
3. Different approximations are used (e.g., z vs. t when n is small).
4. Model assumptions are violated (e.g., nonrandom sample, strong skewness).

Key takeaway:

When both procedures use the same α level and assumptions, the CI and the test will always lead to the same conclusion.